

AI Terminology Glossary

A structured reference of core Artificial Intelligence, Machine Learning, NLP, and MLOps terms — compiled for the Wattly project documentation.

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1. Core Machine Learning Concepts

Artificial Intelligence (AI)

The broad field of building systems that perform tasks normally requiring human intelligence, such as reasoning, perception, and decision-making.

Machine Learning (ML)

A subset of AI in which systems learn patterns from data rather than being explicitly programmed with rules.

Supervised Learning

A training approach where a model learns from labeled input-output pairs to predict outputs for new inputs.

Unsupervised Learning

A training approach where a model finds structure or patterns in data that has no labeled outputs, such as clustering.

Semi-Supervised Learning

A training approach that uses a small amount of labeled data together with a larger pool of unlabeled data.

Reinforcement Learning (RL)

A training approach where an agent learns to make decisions by receiving rewards or penalties from its environment.

Feature

An individual measurable property or input variable used by a model to make predictions.

Feature Engineering

The process of selecting, transforming, or creating input variables to improve model performance.

Label

The known output or target value associated with an input example during supervised training.

Training Set

The portion of a dataset used to fit a model's parameters.

Validation Set

A held-out portion of data used to tune hyperparameters and check for overfitting during training.

Test Set

A separate, unseen portion of data used to evaluate a model's final performance.

Overfitting

When a model learns the training data too closely, including its noise, and performs poorly on new data.

Underfitting

When a model is too simple to capture the underlying pattern in the data, resulting in poor performance overall.

Bias-Variance Tradeoff

The balance between a model being too simple (high bias) and too sensitive to training data (high variance).

Regularization

Techniques such as L1/L2 penalties that discourage overly complex models to reduce overfitting.

Hyperparameter

A configuration value set before training, such as learning rate or number of layers, as opposed to a learned parameter.

Cross-Validation

A technique for assessing model performance by repeatedly splitting data into different train/test partitions.

Ensemble Learning

Combining predictions from multiple models to produce a more accurate or robust result.

Decision Tree

A model that makes predictions by following a tree of if-else conditions based on feature values.

Random Forest

An ensemble of many decision trees whose predictions are averaged or voted on to improve accuracy.

Isolation Forest

An unsupervised ensemble algorithm that detects anomalies by isolating data points through random partitioning; anomalies require fewer splits to isolate.

Gradient Boosting

An ensemble technique that builds models sequentially, each one correcting the errors of the previous ones.

Support Vector Machine (SVM)

A supervised model that finds the optimal boundary (hyperplane) separating classes in feature space.

K-Means Clustering

An unsupervised algorithm that groups data points into K clusters based on similarity.

Dimensionality Reduction

Techniques such as PCA that reduce the number of input variables while preserving important information.

2. Deep Learning & Neural Networks

Neural Network

A model composed of layers of interconnected nodes (neurons) loosely inspired by the brain, used to learn complex patterns.

Deep Learning

A subfield of machine learning using neural networks with many layers to learn hierarchical representations of data.

Perceptron

The simplest type of neural network unit, computing a weighted sum of inputs passed through an activation function.

Activation Function

A function (e.g., ReLU, sigmoid, tanh) applied to a neuron's output to introduce non-linearity into the network.

Weights

Learnable parameters that scale the influence of inputs on a neuron's output.

Bias (in neural nets)

A learnable offset added to a neuron's weighted input sum, distinct from statistical bias.

Backpropagation

The algorithm used to compute gradients and update a neural network's weights by propagating error backward through layers.

Gradient Descent

An optimization algorithm that iteratively adjusts model parameters in the direction that reduces error.

Learning Rate

A hyperparameter controlling how large a step gradient descent takes when updating weights.

Epoch

One complete pass of the training dataset through a model during training.

Batch Size

The number of training examples processed together before the model's weights are updated.

Loss Function

A function that measures the difference between a model's predictions and the true values, guiding training.

Convolutional Neural Network (CNN)

A neural network architecture specialized for grid-like data such as images, using convolutional filters to detect spatial patterns.

Recurrent Neural Network (RNN)

A neural network architecture designed for sequential data, where outputs depend on previous steps in the sequence.

Long Short-Term Memory (LSTM)

A type of RNN with gating mechanisms that allow it to retain information over long sequences, commonly used for time-series forecasting.

Gated Recurrent Unit (GRU)

A simplified variant of LSTM with fewer gates, often faster to train while retaining similar sequence-modeling ability.

Transformer

A neural network architecture based on self-attention that processes sequences in parallel rather than step-by-step, underlying most modern LLMs.

Attention Mechanism

A technique that lets a model weigh the relevance of different parts of the input when producing each part of the output.

Self-Attention

An attention mechanism where a sequence attends to itself, letting each element consider all others when forming its representation.

Embedding

A dense numerical vector representation of data (words, sentences, images) that captures semantic meaning in a lower-dimensional space.

Dropout

A regularization technique that randomly disables neurons during training to reduce overfitting.

Vanishing Gradient

A problem where gradients become extremely small in deep networks, slowing or halting learning in early layers.

Transfer Learning

Reusing a model trained on one task as the starting point for training on a related task.

Fine-Tuning

Further training a pre-trained model on a smaller, task-specific dataset to adapt it to a new use case.

3. Natural Language Processing & Large Language Models

Natural Language Processing (NLP)

The field of AI focused on enabling machines to understand, interpret, and generate human language.

Large Language Model (LLM)

A neural network, typically transformer-based, trained on massive text corpora to understand and generate natural language.

Tokenization

The process of breaking text into smaller units (tokens) such as words or sub-words for model processing.

Token

A unit of text, such as a word or part of a word, that a language model processes as input or output.

Prompt

The input text given to a language model to elicit a response.

Prompt Engineering

The practice of designing and refining prompts to reliably produce desired outputs from a language model.

Context Window

The maximum amount of text (measured in tokens) a language model can consider at once when generating a response.

Hallucination

When a language model generates plausible-sounding but factually incorrect or fabricated information.

Zero-Shot Learning

A model's ability to perform a task it was never explicitly trained on, using only a natural-language instruction.

Few-Shot Learning

Providing a model with a small number of examples within the prompt to guide it toward the desired task behavior.

Temperature (sampling)

A parameter controlling the randomness of a language model's output; higher values produce more varied text.

System Prompt

An instruction given to a language model that sets its behavior, role, or constraints before user interaction begins.

Named Entity Recognition (NER)

An NLP task that identifies and classifies entities such as names, dates, or locations in text.

Sentiment Analysis

An NLP task that determines the emotional tone (positive, negative, neutral) expressed in text.

Word Embedding

A vector representation of words such that semantically similar words are located close together in vector space.

BERT

A transformer-based language model designed to understand context bidirectionally, commonly used for classification and comprehension tasks.

GPT (Generative Pre-trained Transformer)

A family of transformer-based language models trained to generate text by predicting the next token in a sequence.

Gemini

Google's family of multimodal large language models, accessible via API for tasks such as text generation and reasoning.

Multimodal Model

A model capable of processing and generating more than one type of data, such as text, images, and audio together.

Chatbot

A conversational application, often powered by an LLM, designed to interact with users in natural language.

4. Retrieval-Augmented Generation (RAG) & Knowledge Grounding

Retrieval-Augmented Generation (RAG)

An architecture that combines a language model with an external knowledge retrieval step, grounding generated answers in retrieved documents or data.

Knowledge Base

A structured or semi-structured collection of information that a system can query or retrieve from.

Vector Database

A database optimized for storing and searching embeddings by similarity, commonly used to power retrieval in RAG systems.

Semantic Search

A search method that matches meaning rather than exact keywords, typically using embeddings and similarity metrics.

Cosine Similarity

A metric measuring the angle between two vectors, commonly used to compare embeddings for semantic similarity.

Chunking

Splitting large documents into smaller segments so they can be embedded and retrieved effectively in a RAG pipeline.

Grounding

Anchoring a model's output in verified external data or facts to reduce hallucination and improve accuracy.

Retriever

The component in a RAG system responsible for fetching relevant documents or data given a query.

Reranking

A step in retrieval pipelines that reorders initially retrieved results by relevance before passing them to the generator.

Digital Twin

A virtual, data-driven representation of a physical system (e.g., a household's electricity usage) used for simulation or querying.

Knowledge Graph

A structured representation of entities and their relationships, often used to support reasoning and retrieval.

5. Time-Series Forecasting & Anomaly Detection

Time-Series Data

A sequence of data points indexed in time order, such as hourly electricity consumption readings.

Forecasting

The task of predicting future values in a time series based on historical patterns.

Seasonality

Regular, predictable patterns in time-series data that repeat over fixed intervals, such as daily or monthly cycles.

Trend

The long-term upward or downward movement in a time series, independent of short-term fluctuations.

Sliding Window

A technique that uses a fixed-size moving segment of past data points as input for forecasting the next value.

Anomaly Detection

The process of identifying data points or patterns that deviate significantly from expected behavior.

Outlier

A data point that differs significantly from the rest of the dataset, potentially indicating an error or anomaly.

Autoregressive Model (AR)

A statistical model that predicts future values based on a linear combination of past values in the same series.

ARIMA

A classical statistical forecasting method combining autoregression, differencing, and moving averages.

Mean Absolute Error (MAE)

A metric measuring the average magnitude of errors between predicted and actual values, without considering direction.

Root Mean Squared Error (RMSE)

A metric measuring the square root of the average squared differences between predicted and actual values, penalizing larger errors more.

6. Model Evaluation & Metrics

Accuracy

The proportion of correct predictions out of all predictions made by a model.

Precision

Of all instances predicted as positive, the proportion that are actually positive.

Recall

Of all actual positive instances, the proportion that the model correctly identified.

F1 Score

The harmonic mean of precision and recall, balancing both metrics into a single score.

Confusion Matrix

A table summarizing a classification model's predictions against actual labels, showing true/false positives and negatives.

ROC Curve

A graph showing the tradeoff between true positive rate and false positive rate at different classification thresholds.

AUC (Area Under Curve)

A single metric summarizing a model's ability to distinguish between classes across all thresholds, derived from the ROC curve.

Confidence Score

A model's estimated certainty about a given prediction, often expressed as a probability.

Ground Truth

The verified, real-world correct answer or label used to evaluate a model's predictions.

Benchmark

A standard dataset or task used to compare the performance of different models under consistent conditions.

7. MLOps, Deployment & System Architecture

Model Training

The process of fitting a machine learning model's parameters to data.

Inference

The process of using a trained model to make predictions on new, unseen data.

API (Application Programming Interface)

A defined interface that lets software components, such as an Express backend and an AI model, communicate with each other.

Microservice

A small, independently deployable service responsible for a specific piece of functionality within a larger system.

Model Serving

Making a trained model available to receive inference requests, typically through an API endpoint.

Batch Processing

Running computations, such as model inference, on accumulated data at scheduled intervals rather than in real time.

Real-Time Inference

Producing model predictions immediately in response to a request, as opposed to on a delayed schedule.

MLOps

Practices and tooling for reliably deploying, monitoring, and maintaining machine learning models in production.

Model Drift

A decline in model performance over time as real-world data patterns diverge from the data the model was trained on.

Latency

The time delay between a request being made and a response being returned by a system or model.

Scalability

A system's ability to handle increasing amounts of work or data without a proportional loss in performance.

Containerization

Packaging an application and its dependencies into a portable unit (e.g., Docker) that runs consistently across environments.

8. AI Ethics, Safety & Responsible AI

AI Bias

Systematic errors in a model's outputs that unfairly favor or disadvantage certain groups, often stemming from biased training data.

Explainability (XAI)

The degree to which a model's internal decision-making process can be understood by humans.

Interpretability

The extent to which a human can understand the cause of a model's decision or prediction.

Data Privacy

Protecting personal or sensitive information collected and processed by AI systems from misuse or unauthorized access.

Fairness

The principle that an AI system's outcomes should not unjustly disadvantage individuals or groups.

AI Alignment

The effort to ensure an AI system's behavior matches human intentions and values.

Responsible AI

A set of practices aimed at developing and deploying AI systems that are safe, fair, transparent, and accountable.

Adversarial Attack

An input specifically crafted to deceive a model into making an incorrect prediction.